

Measure-Theoretic Foundations and Fourier Pricing of the Heston Model

Quantitative Research Portfolio

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Abstract

This document outlines the rigorous mathematical foundations of the Heston (1993) stochastic volatility model. We establish the physical dynamics of the asset, apply Girsanov's Theorem to construct the equivalent martingale measure, and utilize the Feynman-Kac theorem to derive the pricing Partial Differential Equation (PDE). We then solve this PDE in the Fourier domain using characteristic functions, implementing the Albrecher (2007) branch-cut fix, and demonstrate how the model's parameters mathematically generate the empirical implied volatility smile observed in equity markets.

1 The Physical Measure $\mathbb{P}$ and Girsanov's Theorem

We define a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \in [0, T]}, \mathbb{P})$. Under the historical measure $\mathbb{P}$, the asset price S_t and its instantaneous variance V_t follow a coupled system of Stochastic Differential Equations (SDEs):

$$dS_t = \mu S_t dt + \sqrt{V_t} S_t dW_t^S \quad (1)$$

$$dV_t = \kappa(\theta - V_t)dt + \sigma\sqrt{V_t}dW_t^V \quad (2)$$

where $d\langle W^S, W^V \rangle_t = \rho dt$. To preclude arbitrage, we require an Equivalent Martingale Measure (EMM), denoted $\mathbb{Q}$. We define the Radon-Nikodym derivative using the market prices of risk $\lambda_S(t, S_t, V_t) = \frac{\mu - r}{\sqrt{V_t}}$ and $\lambda_V(t, S_t, V_t) = \lambda\sqrt{V_t}$.

By Girsanov's Theorem, we shift the drift of our Brownian motions to yield the risk-neutral dynamics under $\mathbb{Q}$:

$$dS_t = rS_t dt + \sqrt{V_t} S_t d\tilde{W}_t^S \quad (3)$$

$$dV_t = \kappa^*(\theta^* - V_t)dt + \sigma\sqrt{V_t}d\tilde{W}_t^V \quad (4)$$

where $\kappa^* = \kappa + \lambda\sigma$ and $\theta^* = \frac{\kappa\theta}{\kappa + \lambda\sigma}$. For notational simplicity in subsequent sections, we drop the asterisks.

2 Feynman-Kac and the Heston PDE

Let $U(t, S, V)$ be the price of a European call option. By risk-neutral valuation, $U(t, S, V) = \mathbb{E}^{\mathbb{Q}}[e^{-r(T-t)}(S_T - K)^+ \mid \mathcal{F}_t]$. Because the discounted option process $e^{-rt}U(t, S_t, V_t)$ is a

$\mathbb{Q}$ -martingale, setting the drift of its Itô expansion to zero yields the fundamental Heston PDE:

$$\frac{\partial U}{\partial t} + \frac{1}{2}V S^2 \frac{\partial^2 U}{\partial S^2} + \rho\sigma V S \frac{\partial^2 U}{\partial S \partial V} + \frac{1}{2}\sigma^2 V \frac{\partial^2 U}{\partial V^2} + rS \frac{\partial U}{\partial S} + \kappa(\theta - V) \frac{\partial U}{\partial V} - rU = 0 \quad (5)$$

3 Fourier Inversion and the Characteristic Function

Solving the 2D PDE numerically via finite differences is computationally prohibitive for calibration. Instead, we move to the log-asset domain $x_t = \ln(S_t)$ and evaluate the characteristic function of the terminal log-price:

$$\phi(u, x, V, \tau) = \mathbb{E}^{\mathbb{Q}} [e^{iux\tau} \mid x_t = x, V_t = V] \quad (6)$$

By the Feynman-Kac theorem, ϕ satisfies a PDE similar to the pricing equation but with boundary condition $\phi(u, x, V, 0) = e^{iux}$. Heston postulates an affine solution of the form:

$$\phi(u, x, V, \tau) = \exp(C(u, \tau) + D(u, \tau)V + iux) \quad (7)$$

Substituting this into the PDE separates it into two Riccati ordinary differential equations. To prevent numerical instability caused by the complex logarithm crossing the principal branch (the "branch cut" problem), we utilize the Albrecher (2007) formulation for $C(u, \tau)$:

$$C(u, \tau) = riu\tau + \frac{\kappa\theta}{\sigma^2} \left((b - \rho\sigma iu - d)\tau - 2 \ln \left(\frac{1 - ge^{-d\tau}}{1 - g} \right) \right) \quad (8)$$

where $d = \sqrt{(\rho\sigma iu - b)^2 - \sigma^2(-u^2 - iu)}$ and $g = \frac{b - \rho\sigma iu - d}{b - \rho\sigma iu + d}$.

The European call price is then recovered via Gil-Pelaez inversion:

$$C(S, K, \tau) = S_t P_1 - K e^{-r\tau} P_2 \quad (9)$$

where $P_j = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \text{Re} \left(\frac{e^{-iu \ln(K) \phi_j(u)}}{iu} \right) du$.

4 Theoretical Justification of the Volatility Smile

The standard Black-Scholes model assumes constant volatility, which implies a flat Implied Volatility (IV) surface. Empirically, equity options exhibit a pronounced "smirk" or "smile." The Heston model organically generates this surface through its parameters:

- **The Leverage Effect** ($\rho < 0$): A strong negative correlation between asset returns and variance produces a negatively skewed return distribution. As the market drops, volatility spikes, increasing the value of out-of-the-money (OTM) puts relative to OTM calls. This tilts the smile into a downward-sloping smirk.
- **Volatility of Volatility** ($\sigma > 0$): The presence of stochastic volatility introduces kurtosis (fat tails) into the return distribution. Extreme market movements are more likely than under a log-normal distribution, increasing the price of all deep OTM options and generating the convexity (the U-shape) of the smile.

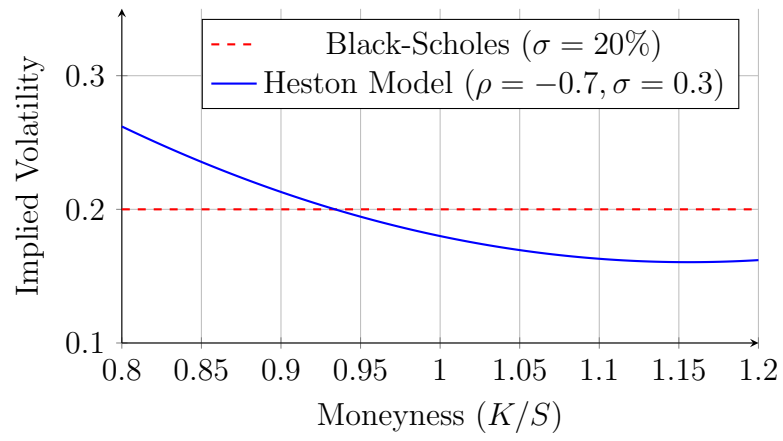


Figure 1: Theoretical comparison of constant volatility versus a Heston-generated volatility smirk, capturing the empirical market demand for downside protection.